An Alignment Check for Debiasing Corrections: A Negative-Control Case Study in Drug Repurposing

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Abstract. A bias-diagnostic metric can judge a debiasing correction only if it is sensitive to what that correction changes; two corrections with different mechanisms can target different quantities, and a metric tuned to one will misjudge the other regardless of which is substantively better. We formalize this as an alignment check: score each correction's own penalty term as a standalone classifier of whatever the bias metric flags, and treat near-chance AUROC as evidence the metric cannot see that mechanism at all. We develop the check through a case study in drug repurposing, where a model trained to rank candidates for one indication also learns what is shared across drugs on the same organ system; in our data the convulsant bemegride ranks first among non-antiepileptic candidates for epilepsy. We test two corrections for this confound, naive area-score subtraction and a negative-control method adapted from genomics and pharmacoepidemiology, across depression, antipsychotic, and epilepsy rankings. The naive correction wins on every bias-diagnostic metric we measured, which looks like a negative result for the principled method until the alignment check is applied: its penalty term carries essentially no information (0.50 AUROC) about the compounds our bias metrics flag, while naive subtraction's term separates them well above chance (0.60–0.79). A shrinkage follow-up, a calibration check, and a direct test of the method's own identifiability assumption rule out noise and misspecification as explanations. We then close the loop formally: the correction's penalty term is, by construction, exactly orthogonal to the representation its base model is fit on, and we confirm empirically that the bias metrics' flags are themselves predictable from that same representation (0.51–0.72 AUROC). Together these explain the result without appeal to this dataset's specifics, and the same two-step check, orthogonality by construction plus empirical flag-predictability, applies to any comparison between a residualized correction and a membership-based bias metric.

1 Introduction

A bias-diagnostic metric is itself a measurement, and a measurement has a detection limit. When two debiasing corrections are compared against one bias metric and one wins, the usual reading is that the winning correction removes the bias

better. A second reading is available whenever the two corrections work through different mechanisms: the metric may simply be built to see one mechanism and not the other, in which case the comparison never tested what it looked like it was testing. Telling these readings apart requires checking, directly, whether the metric could have detected the losing correction's effect at all, rather than inferring quality from the scoreboard alone. We call this an alignment check: score each correction's own penalty term, the quantity it subtracts or projects out to produce its debiased score, as a standalone classifier of whatever the metric flags. A term that classifies the flag near chance is one the metric cannot reward or penalize, independent of whether the correction behind it is good.

We develop this check through a case study chosen because it produces exactly the puzzle the check is built to resolve. Chemical structure is not the only thing a repurposing model can learn to exploit. A model trained to separate antiepileptic drugs from everything else also, incidentally, learns what distinguishes neurology drugs from the rest of the pharmacopeia, and that second signal rides along with the first in every prediction. The practical consequence shows up at the top of a ranked candidate list: in a plain ECFP4 fingerprint model trained on known antiepileptic drugs, bemegride, a convulsant once used to induce seizures for electroconvulsive therapy, ranks first among 1,522 non-antiepileptic candidates, ahead of every genuinely plausible repurposing lead. We reproduce this finding exactly (score 0.7942, matching a prior run to four decimal places) as the motivating case for this paper. A companion measurement study from this group audited evaluation-protocol choices in LINCS-based repurposing models and reported this same ranking as part of a broader audit of distinct evaluation defects [11]; that study proposes no new model and explicitly leaves correction methods to future work.

The same pattern appears in aggregate, not in one example alone. Restricting to drugs with a WHO ATC classification, candidates flagged as nervous-system drugs make up 18.8% of the top 50 false positives for a depression model and 15.2% for an antipsychotic model, against base rates of 10.2% and 10.6%. The enrichment, roughly 1.5 to 1.8 times base rate, is smaller than a name-matched Drug Repurposing Hub label had suggested; ATC covers only 9.7% of the LINCS library, mostly marketed drugs, so both numbers come from a narrower, more defensible population than before. We use the ATC-based numbers throughout.

Two corrections suggest themselves for this confound. The simple one subtracts a generic area score from the disease-specific score, directly penalizing membership in the confounded class; prior informal analysis of this group's work assumed this collapses precision too far to be usable. The more principled one borrows an idea from genomics and pharmacoepidemiology: estimate the shared organ-system factor from negative-control indications, diseases in the same area treated by a different mechanism, and remove only that factor rather than a blunt membership score. We built and evaluated the principled version. It did not beat the simple one on any bias-diagnostic metric we measured, which is where this paper could have stopped and reported a negative result.

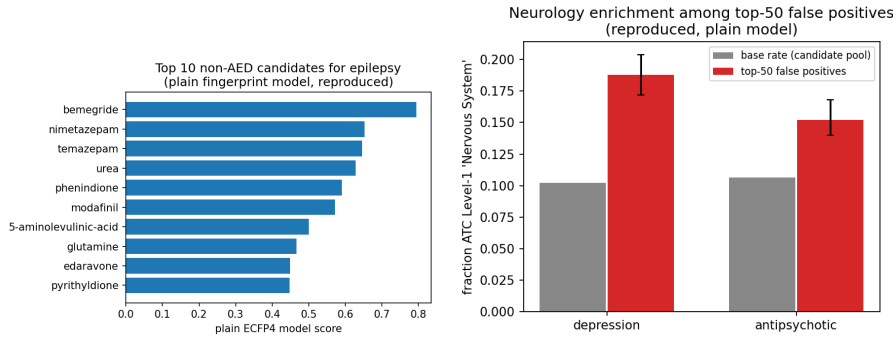


Fig. 1. Left: the top 10 non-antiepileptic candidates ranked by a plain ECFP4 fingerprint model trained on known antiepileptic drugs; bemegride, a convulsant, ranks first. Right: nervous-system drugs (ATC Level-1 'N') are enriched among the top-50 false positives of depression and antipsychotic models relative to each candidate pool's base rate.

Instead we apply the alignment check, and it changes the conclusion. Section 4 proves the principled correction's penalty term is, by construction, exactly orthogonal to the representation its base model is fit on. Section 5.7 shows the bias metrics' flags do not live in that orthogonal complement, they are predictable from the same base representation, while naive's penalty term carries no such constraint, which is why it tracks the flags well. Table 2's comparison is real, but it is evidence this metric cannot see the variation the principled method removes, not that the method fails to remove it.

2 Related Work

Negative controls for unwanted variation. Removing unwanted variation (RUV) using negative controls is an established genomics technique: RUV-2 [7] estimates latent confounding factors from known-null control genes and removes them before testing; later work [8] unifies several RUV variants under one factor-analysis framework. Surrogate variable analysis [14] and empirical-Bayes batch correction [10] address related but distinct versions of the problem; ComBat needs known batch labels rather than negative controls, which is why we follow the negative-control strategy instead. Negative-control exposures and outcomes in pharmacoepidemiology [16] formalize the same logic for observational drug-effect confounding and remain in active use [21]. Our area-factor step is a direct structural analogue, applied to indications rather than genes or exposures; the statistical idea is not new, only its application to cross-indication repurposing confounding.

Debiasing drug-repurposing and DTI models. NRBdMF [1] factorizes a signed drug×disease matrix from SIDER [13], indications labeled -1 and side effects $+1$, to reduce false positives through *within-drug* directionality; it never uses

other diseases as negative controls or removes a shared area factor, so we include it as a baseline. TAPB [15] and DRDM [17] address other biases in drug-target/drug-repositioning prediction (target-popularity and graph-degree bias); neither targets organ-system confounding or uses negative-control indications. SIDER-derived phenotypic similarity was first shown to carry repositioning-relevant signal by Campillos et al. [3,12], the basis for our hard negatives.

Benchmarks and resources. repoDB [2] provides a curated gold-standard set of approved and failed drug-disease pairs and is a standard secondary comparison point in this literature. The Drug Repurposing Hub [5], source of a name-matched therapeutic-area label used in prior work from this group, has limited coverage; we replace it with ATC classification. Epileptogenic/proconvulsant drug lists have an established clinical literature base [18,4,19], used to validate the hard-negative panel for epilepsy (Section 3).

Validity of bias-evaluation metrics. A smaller literature asks not whether a method beats a baseline but whether the protocol can detect what it claims to measure: scaffold-split virtual-screening benchmarks overestimate performance relative to a realistic split [9], and affinity benchmarks reward memorization of training-set similarity unless reweighted against it [23]. Neither originates in computational biology, and we know of no close RECOMB precedent. Our alignment check sits in this family, one level more specific: whether a given bias-diagnostic metric can see a given correction's mechanism at all, stated in Section 4 as a general, checkable precondition.

3 Data

We build on the nine ChEMBL [22] indication classes used in our companion study (epilepsy, depression, antipsychotics, antibacterials, antidiabetics, antineoplastics, lipid-lowering drugs, NSAIDs, betablockers as an antihypertensive proxy), over the same 9,878-compound LINCS library, extended with nine further ATC-defined indications (Parkinson's disease, anxiolytics, antidementia drugs, opioid analgesics, renin-angiotensin antihypertensives, antacids, osteoporosis drugs, antimycotics, immunosuppressants) chosen so every original class has at least one same-area, different-mechanism negative-control partner. ATC Level-1 codes come from the ChEMBL molecule API, by full InChIKey; only 954 of 9,878 compounds (9.7%) carry any ATC classification, since most of the library is research or tool compounds rather than marketed drugs, the main reason the enrichment statistics in Section 5.1 are smaller than a name-matched Hub label had suggested.

SIDER [13] side-effect data is joined to the full compound set by PubChem CID, extending a prior join that covered only 1,725 compounds. Coverage remains low, 543 of 9,878 (5.5%), for the same reason as ATC: SIDER only covers marketed drugs with a package insert. We use its characteristic adverse-effect

terms as hard negatives, convulsion-family terms for epilepsy, hypotension for antihypertensives, hyperglycaemia for antidiabetics, for classes where such a term map makes sense, without inventing one where it does not.

For epilepsy specifically, we use a 43-drug proconvulsant list from this group's own prior, unpublished work, stated plainly rather than treated as an external resource: literature-curated from Chen, Albertson & Olson [4], a peer-reviewed clinical review of drug-induced seizures, and cross-validated against two public sources. Of 82 canonical compounds extracted from the original curation, 62 resolve to a compound in our library; of those, 38 (61%) carry a SIDER-listed convulsion or seizure adverse effect, and 24 (39%) match a risk tier in the independently published systematic review of Ruffmann, Bogliun & Beghi [18]. The SIDER overlap is substantial given SIDER's own sparse 5.5% coverage, since most proconvulsants on this list are marketed drugs that resolve to SIDER at a much higher rate than the library average.

4 Method

Each compound is represented by concatenating a 32-component PCA of its ECFP4 fingerprint (TruncatedSVD, chosen to avoid indication-matrix leakage into the chemistry features) with a 16-dimensional embedding from a free logistic matrix factorization fit jointly across all 18 indications. Chemistry is necessary rather than cosmetic: a pure compound-by-indication co-occurrence factorization carries almost no signal on this matrix, since 94% of the library has zero positive indications across all 18 classes and per-indication AUPRC on training data is indistinguishable from base rate. This is why NRLMF and NRBdMF add drug or disease similarity through neighborhood regularization rather than relying on the interaction matrix alone; we add chemistry for the same reason.

On top of this shared representation, we compare four ways of scoring candidates for a target indication k_0. *Plain* fits one logistic regression per indication on the shared representation, with no area correction. *Naive subtraction* scores a candidate as the plain score minus λ times a min-max-normalized mean indication rate over k_0's negative-control set, $\lambda = 2$. *NRBdMF* factorizes a signed drug-by-disease matrix directly, following Azuma et al. [1]: -1 for a ChEMBL/ATC indication, $+1$ for a SIDER adverse-effect hit, chemistry withheld, matching the original specification. *Proposed* is the negative-control correction this paper is actually about: for k_0's negative-control set $\mathrm{NC}(k_0)$, diseases in the same ATC Level-1 area treated through a different mechanism, take the train-only residuals of the already-fit plain models restricted to $\mathrm{NC}(k_0)$'s columns, take the leading left singular vectors of that residual as an area factor A, following the RUV-2 strategy of Gagnon-Bartsch and Speed [7] applied to indications rather than genes, refit k_0 with [chemistry+MF, A], and score with A's coefficients dropped, projecting out whatever the negative controls explain.

We later tested whether a shrunk version of this correction, scaling A by a weight $s \in [0, 1]$ before use, following the unified RUV family of Gerard and Stephens [8], would improve matters; Section 5.5 reports that it does not.

4.1 The alignment check

Any correction scoring a candidate as a base score minus (or plus) a penalty term, naive subtraction's λ-weighted area rate and the proposed method's projected-out A-coefficients both have this form, can be probed independently of whatever metric judges it. Given a bias-diagnostic metric defined through a binary flag F (same-area membership, hard-negative membership, or any other criterion the metric penalizes), the check scores the correction's own penalty term P as a standalone classifier of F via $\mathrm{AUROC}(P, F)$. Near-0.5 AUROC means P carries no information about F: the metric cannot reward or penalize that mechanism regardless of correction quality, so a comparison against a baseline whose term *is* informative about F is not on equal footing. We apply this to our two corrections in Section 5.7; the check names no drug-repurposing-specific quantity.

Section 5.7's result has a structural reason, stated as a proposition about how the proposed method's penalty term is built. **Proposition.** Let X be the shared representation (chemistry PCA concatenated with the free matrix-factorization embedding) on which every negative-control indication's model is fit by maximum-likelihood logistic regression, and $\hat{P}$ that model's fitted probabilities on the negative-control columns, train rows only. Each residual $r_j = y_j - \hat{p}_j$ satisfies the logistic score equation $X_{\mathrm{tr}}^{\top} W r_j = 0$ for the fitting weights W (the standard GLM score equation under canonical link, here class-balanced): r_j is exactly orthogonal to every column of X on train rows, in the W-weighted inner product. A is the leading left singular vectors of $R = [r_1, \ldots, r_{|\mathrm{NC}|}]$, hence a linear combination of its columns, so A inherits this exact orthogonality.

This is algebraic, true regardless of what the controls or target represent, and says nothing by itself about whether F relates to A; that depends on whether F is predictable from X. If it is, the orthogonality explains why A carries little information about F even though raw negative-control membership, naive's penalty term, with no such constraint, can. Section 5.7 tests both halves: whether A behaves as predicted, and whether F is predictable from X.

5 Experiments

5.1 Reproducing the motivating confound

We retrained the plain ECFP4 fingerprint model for epilepsy, depression, and antipsychotics, averaged over five seeds, and confirmed both parts of the motivating observation from Section 1: bemegride ranks first of 1,522 non-antiepileptic candidates at a score matching the pre-existing result to four decimal places, and the top-50 false positives for depression and antipsychotic models are enriched for nervous-system drugs at roughly 1.5 to 1.8 times each pool's base rate (18.8% vs. 10.2%, and 15.2% vs. 10.6%, respectively). We looked for, but could not locate in any saved script in this group's prior work, the specific AUPRC collapse from 0.45 to 0.17 that an earlier informal analysis had attributed to naive area-score subtraction. We report what we can actually measure instead.

Table 1. AUPRC on true indications, CVS2 (mean over 5 reruns; 95% bootstrap CI in parentheses).

Target	Plain	Naive subtract	NRBdMF	Proposed
depression	0.201 (.114,.289)	0.190 (.105,.262)	0.039	0.181 (.112,.247)
antipsychotic	0.440 (.252,.628)	0.412 (.215,.608)	0.039	0.437 (.264,.611)
epilepsy	0.091 (.065,.127)	0.090 (.067,.128)	0.024	0.085 (.065,.112)

Table 2. Bias-diagnostic metrics. Top-50 same-area rate: lower is better (base rate ≈ 0.09–0.11). Hard-negative and proconvulsant rank: higher percentile is better (lower in the candidate list).

Metric / Target	CVS	Plain	Naive subtract	NRBdMF	Proposed
area rate, depression	2	0.112	0.024	0.084	0.112
area rate, antipsychotic	2	0.136	0.096	0.084	0.144
area rate, epilepsy	2	0.144	0.068	0.120	0.176
hard-neg. pct., epilepsy	2	38.4	50.9	47.8	42.0
proconv. pct., epilepsy	2	30.9	71.8	49.2	26.3

5.2 Evaluation protocol

We evaluate under three cross-validation schemes matched to the matrix-factorization repurposing literature: CVS1, a random 20% holdout of matrix cells; CVS2, a cold-start-drug holdout of 20% of compounds; and CVS3, a cold-start-disease holdout of the entire target column. Each scheme runs five times with independent splits and initializations, on an identical 1,702-compound drug pool shared by all four methods. For CVS2 we also report a scaffold-disjoint variant, since a random compound holdout can still leak near-duplicate scaffolds between train and test.

5.3 Main results

Table 1 reports CVS2 AUPRC on the true indication label (CVS1 and CVS3 tables, same pattern, are in the repository); Table 2 reports the two bias-diagnostic metrics that matter most here, the same-ATC-area rate among the top 50 false positives and the median rank percentile of SIDER-flagged hard negatives.

Three things stand out. NRBdMF performs close to chance everywhere (AUPRC 0.02 to 0.05), confirming what the matrix diagnostics already suggested: a pure within-drug signed factorization, no chemistry, has little to work with on an 18-indication matrix this sparse. The proposed method tracks plain closely on AUPRC, the precision-preservation property it was designed for, but on both bias metrics in Table 2 it is dominated by naive subtraction, often by a wide margin; the clearest case is the proconvulsant panel, where naive pushes it from the 31st percentile to the 72nd while proposed moves it under 5 points, and in the epilepsy CVS2 row in the wrong direction. Contrary to this paper's own motivating assumption, naive subtraction's AUPRC cost is modest across every row measured under all three CVS schemes, never approaching the 62% relative collapse an earlier informal analysis had reported.

5.4 Negative-control validity check

A method that identifies an area factor from negative controls should, if those controls are mechanistically independent, leave each control's own predictability roughly unchanged when corrected using a factor estimated from the *other* controls. We ran this leave-one-negative-control-out diagnostic for every (target, control) pair; 13 of 14 cells cluster within ± 0.03 of zero AUPRC change, consistent with independence. The exception, anxiolytic under both antipsychotic and epilepsy targets, showed a large positive delta at 4 to 5 reruns ($+0.127$, $+0.126$) but also the largest standard deviation by a wide margin; re-running with 20 seeds shrank the mean to $+0.034$ with the standard deviation essentially flat, the signature of small-sample noise from anxiolytic's 7-positive class rather than a calibration failure (full per-control plot in the repository).

5.5 Shrinkage follow-up

The most natural explanation for Table 2 is that the negative-control classes are small, 4 to 21 positives each, and the resulting area factor is mostly noise. We tested this with empirical-Bayes-style shrinkage of the factor: $A \leftarrow sA$ for $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$, selecting s and the factor rank via the same leave-one-negative-control-out diagnostic (excluding $s = 0$, which trivially achieves perfect calibration by reducing to plain). The diagnostic selected full-strength, unshrunk correction ($s = 1$) for all three targets, and re-evaluation at that setting was statistically indistinguishable from the original, ruling out noisy estimation. The more likely account is structural: the representation already absorbs much of the organ-system signal a negative-control factor would otherwise capture, or a residual-SVD strategy of this kind does not transfer well to a matrix this sparse. An RUV-4-style joint alternating estimator, re-estimating the factor and controls' fits together over several rounds, did not beat naive subtraction either, so the failure is not specific to two-stage RUV-2.

5.6 Testing the identifiability-violation hypothesis directly

RUV's own identifying assumption [8] is that a negative-control column must carry zero loading on the covariate of interest. If that fails here, the area factor would absorb some of the target's own signal, and projecting it out would cost real precision along with the confound. For each target and CVS scheme where the area factor is fit, we regressed the target's true label, train rows only, onto its fitted area factor, and measured R^2, which should sit near zero under the assumption. Across all six cells the values are small, 0.0006 to 0.021, and do not track naive's advantage: epilepsy has the smallest overlap but the largest gap in naive's favor, the opposite of what the hypothesis predicts. This rules it out.

5.7 Applying the alignment check

Ruling out identifiability leaves a more mundane candidate: the bias metrics may sit closer to what naive subtraction directly penalizes, raw negative-control

Table 3. AUROC of each method's own penalty term, and of the base representation X alone, as standalone classifiers of the compounds the bias metrics flag (mean ± std, 5 seeds, CVS2). 0.5 = no information about the flag.

Target	Flag	Naive's term	Proposed's term	X alone
depression	same-ATC-area	0.766 ± 0.051	0.499 ± 0.002	0.718 ± 0.015
depression	hard-negative	0.636 ± 0.025	0.499 ± 0.002	0.514 ± 0.066
antipsychotic	same-ATC-area	0.774 ± 0.046	0.498 ± 0.005	0.706 ± 0.038
antipsychotic	hard-negative	0.663 ± 0.033	0.501 ± 0.002	0.591 ± 0.065
epilepsy	same-ATC-area	0.787 ± 0.055	0.497 ± 0.007	0.722 ± 0.040
epilepsy	hard-negative	0.600 ± 0.042	0.500 ± 0.002	0.613 ± 0.040
epilepsy	proconvulsant	0.721 ± 0.084	0.500 ± 0.002	0.695 ± 0.081

membership, than to what the proposed method targets, a residualized factor estimated after chemistry and matrix-factorization signal are removed. We apply the alignment check of Section 4.1: for each target, under CVS2, we compute naive's penalty term (raw negative-control rate) and proposed's penalty term made concrete (the contribution A's coefficients would have made had they not been projected out) for every non-indicated test compound, and score each, on its own, as a classifier of the exact compounds the bias metrics flag: same-ATC-area membership, SIDER-hard-negative, or proconvulsant-panel membership.

Table 3 is the resolution, not a further hint. Naive's penalty term separates every bias-flagged compound set well above chance, 0.60 to 0.79 AUROC, across every target. The proposed method's penalty term sits at 0.497 to 0.501 in every single row, with a standard deviation of 0.002 to 0.007, too tight for noise to explain the gap. By the Proposition in Section 4.1, the proposed method's penalty term is exactly orthogonal, on train rows, to the chemistry-plus-matrix-factorization representation X that every per-indication model is fit on; that explains near-chance AUROC against flag F only to the extent F is predictable from X itself. The rightmost column tests that premise rather than assuming it, by fitting a plain classifier on X alone, train rows only, to predict each flag on the same candidate population. The same-ATC-area flag is well predicted by X on every target, 0.706 to 0.722, close to naive subtraction's own 0.766 to 0.787: most of what makes a compound look like it belongs to the confounded area is already present in chemistry and the shared matrix-factorization embedding before any negative-control information is added. The hard-negative and proconvulsant flags are predictable from X too, but less uniformly, from 0.514 for depression's hard negatives (indistinguishable from chance) up to 0.695 for the epilepsy proconvulsant panel. Together with the Proposition, this account does not rest on this dataset's specific numbers. The same-ATC-area row is the clean case; the hard-negative row is a real but weaker instance of the same pattern.

This two-step recipe, orthogonality by construction, then an empirical check of whether the metric's flag is predictable from the correction's base representation, names nothing specific to drug repurposing, ATC codes, or RUV, and applies wherever a residualized correction meets a bias metric built on group membership the same base features already predict.

We also re-ran this check outside drug repurposing, on the dataset RUV-2 was introduced on: Vawter et al.'s brain gene-expression study [20], gender as the factor of interest, lab as the batch confound, and Gagnon-Bartsch and Speed's 799 housekeeping negative controls [6]. Step one holds exactly: each control gene's OLS residual on gender satisfies the normal equations to floating-point precision, so the genomics analog of A is exactly orthogonal to gender. Step two differs: lab is not predictable from gender (AUROC 0.39–0.41, near or below chance), unlike same-ATC-area membership above. As predicted, the orthogonalized factor stays informative about lab (0.50–0.94 alone, 0.95–1.00 combined), matching or beating naive's unprojected control-gene summary (0.55–0.76): the two domains diverge exactly where the check says they should. Donor-grouped cross-validation leaves these numbers unchanged.

6 Discussion and Limitations

This paper set out to build a more principled alternative to a correction assumed too costly to use, found instead that naive wins on every bias metric measured, and asked whether that scoreboard meant what it appeared to. It did not: the Proposition in Section 4.1 shows the principled correction's penalty term is exactly orthogonal to its base representation by construction, and Table 3's rightmost column shows the flags are substantially predictable from that same representation. A term built to contain none of what the representation explains has little left to say about a flag it already explains well. This is not a post-hoc reinterpretation: Sections 5.6 and 5.7 ruled out an identifiability violation first, before we confirmed the alignment account.

We think the check itself, independent of which correction wins here, is the more durable contribution: negative-control identification of a latent confound is well established in genomics [7,8,14] and pharmacoepidemiology [16,21], and a researcher applying it elsewhere will often compare it against a simpler baseline on whatever bias metric is at hand, exactly the setting that misleads if the flag aligns more with the baseline's term. Section 5.7's recipe catches that before it is reported as a finding about which correction is better. NRBdMF [1] is not designed to remove organ-system confounding, so its weak performance comments on bare co-occurrence signal, not the general approach; it also has no residualized penalty term, scoring candidates directly from a signed factorization. Not every method decomposes this way, and knowing which do is part of using the check correctly.

Several limitations remain. The check now spans two domains (Section 5.7); its empirical half must be re-run for each new case, unlike the orthogonality half. ATC and SIDER cover under 10% of the LINCS library, capping how strong an ATC- or SIDER-grounded correction can be measured. CVS3 is degenerate for methods needing target-specific signal: plain and proposed collapse to the same transfer score. Table 3 measures each term against the bias metrics' own flags, not ground truth. We skipped repoDB [2] and did not read DRDM [17] in full. Code: `https://github.com/edanmn/area-debiasing-drug-repurposing`.

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